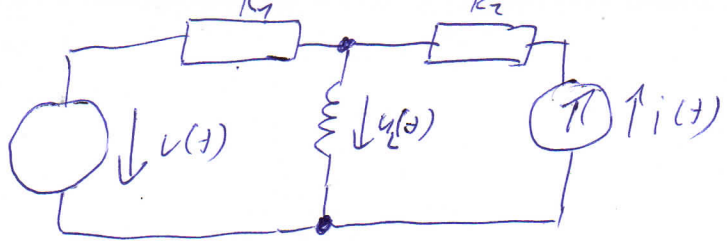


$u(t) = 10 \sin(1000\, t)\ [V]$

$i(t) = 10 \sin(1000\, t - \frac{\pi}{2})\ [mA]$

$R_1 = 1\ k\Omega$

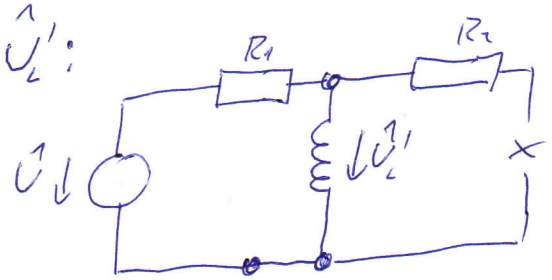
$R_2 = 1\ k\Omega$

$L = 1\ H$

$\hat{U} = 10 e^{j \cdot 0} = 10\ [V]$

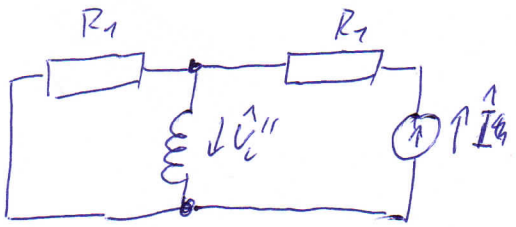
$\hat{I} = 10 e^{-j\frac{\pi}{2}} = -10j\ [mA]$

$\hat{U}_L'$:



$$\hat{U}_L' = \hat{U} \cdot \frac{j\omega L}{j\omega L + R_1} = 10[V] \cdot \frac{1000j[\Omega]}{(1000j + 1000)[\Omega]} = (5 + 5j)[V]$$

$\hat{U}_L''$:



$$\hat{U}_L'' = \hat{I} \cdot \frac{j\omega L\ R_1}{R_1 + j\omega L} = 10 e^{-j\frac{\pi}{2}} [mA] \cdot \frac{1000j[\Omega] \cdot 1000[\Omega]}{(1000 + 1000j)[\Omega]} = (5 - 5j)[V]$$

$\hat{U}_L = 10\ [V]$

$u_L(t) = 10 \sin(1000\, t)\ [V]$